

Fig. 3.1 shows a mass-spring system placed on a frictionless slope. The slope has an angle of θ from the horizontal. When a block of mass m is hung, the spring stretches by an extension of e and the mass remains in equilibrium. The spring is further extended by x downwards, along the slope, and released for the mass-spring system to oscillate. The spring constant is k .

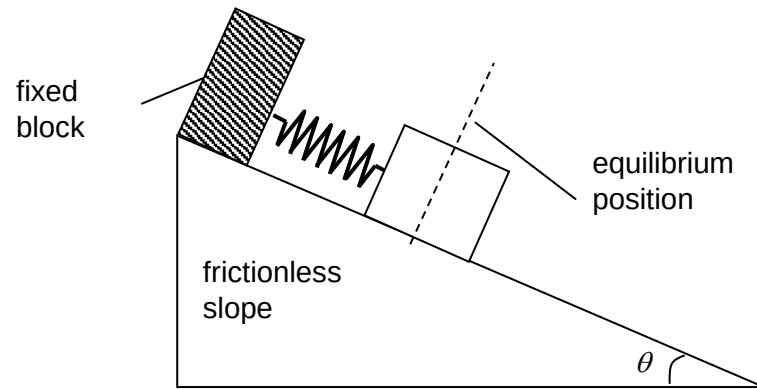


Fig. 3.1

- (a) By using Newton's second law, show that the acceleration a of the block at the lowest point is given by $a = -\frac{k}{m}x$.

[2]

- (b) The amplitude of oscillation of the mass-spring system is 3.0 cm. Calculate the position of the mass from equilibrium when the speed of the mass is 25 % of the maximum speed.

position = cm [2]

[Turn over]

- (c) A student removes the fixed block and attaches a variable frequency oscillator to the mass-spring system, as shown in Fig. 3.2.

Fig. 3.3 shows the variation of the amplitude of mass with the frequency of the oscillator.

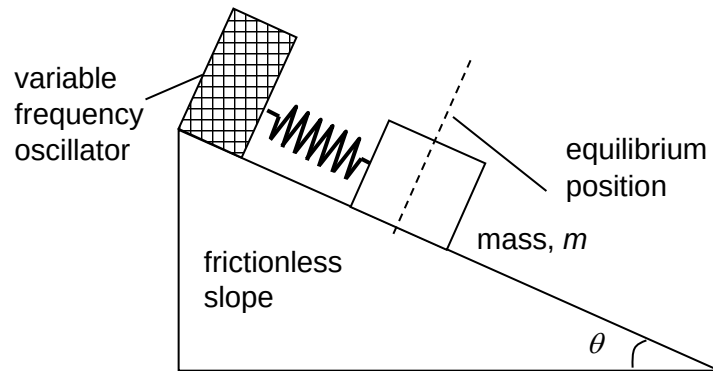


Fig. 3.2

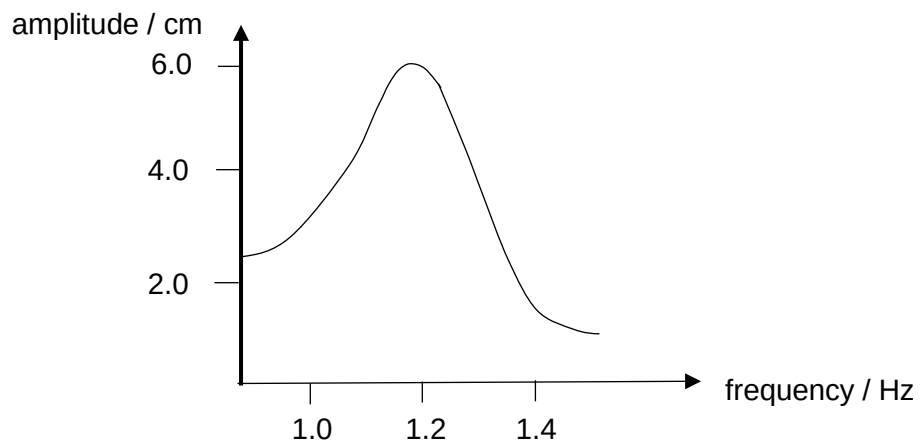


Fig. 3.3

- (i) Explain the phenomenon illustrated in Fig. 3.3.

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..... [2]

- (ii) Using Fig. 3.3 calculate the magnitude of maximum acceleration of the mass.

acceleration = m s^{-2} [2]

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[4 (a) State Coulomb's law between 2 point charges]